Graphing Rational Functions

Name _____

Algebra 2 Lesson 4-2 (fast track) Date _____ Period _____

Learning goal: I can find the vertical and horizontal asymptotes of a rational function and use them to graph it.

Language goal: I can explain how I found each asymptote using the words numerator, denominator, degree, asymptote, and leading coefficient.

On the quiz: (1) Bonus, +5: match a graph to its equation by its two asymptotes and one point (Example 4). (2) A factor that cancels is not an asymptote (full Savvas quiz only). (3) Graded, 10 points: find both asymptotes of a linear-over-linear function (Example 4). (4) Water added to a mixture (full Savvas quiz only); the Practice block is optional preparation for it. (5) Graded, 10 points: pick the function with a given horizontal asymptote by comparing degrees and leading coefficients (Example 3).

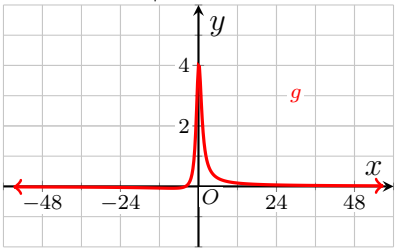
Example 3: Find Types of Horizontal Asymptotes

What are the horizontal asymptotes for the graph of $f(x) = \dfrac{3x-2}{x^2+7x+12}$?

A horizontal asymptote describes **end behavior**: what y does as $x \to \infty$ or $x \to -\infty$. Compare the **degree** (highest power) of the **numerator** with the degree of the **denominator**. There are three cases.

Case 1: degree of numerator < degree of denominator. Horizontal asymptote $y = 0$.

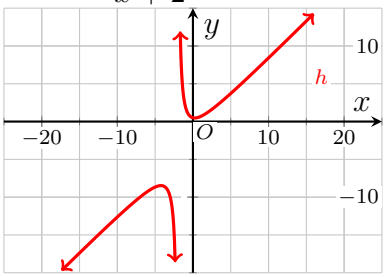
$g(x) = \dfrac{x+4}{x^2+1}$ $\quad (1 < 2)$



The bottom grows faster, so $g(x) \to 0$.

Case 2: degree of numerator > degree of denominator. **No** horizontal asymptote.

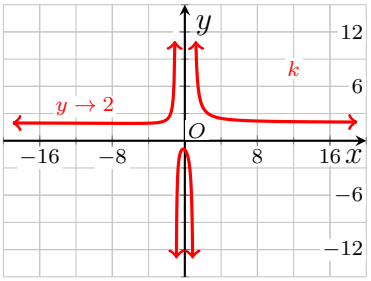
$h(x) = \dfrac{x^2+1}{x+2}$ $\quad (2 > 1)$



The top grows faster, so $y \to \pm\infty$.

Case 3: degrees equal. Horizontal asymptote $y =$ the ratio of the **leading coefficients**.

$k(x) = \dfrac{2x^2+x+1}{x^2-1}$ $\quad (2 = 2)$



Long division: $k(x) = 2 + \dfrac{x+3}{x^2-1}$, so $y = \frac{2}{1} = 2$.

Back to f. $f(x) = \dfrac{3x-2}{x^2+7x+12}$: the numerator has degree 1 and the denominator has degree 2. That is Case 1, so the horizontal asymptote is $\boldsymbol{y = 0}$.

Say it: The degree of the numerator of f is _____ and the degree of the denominator is _____, so the horizontal asymptote is _____ because …

Try It 3: Horizontal asymptotes

What are the horizontal asymptotes of the graph of each function?

a. $g(x) = \dfrac{2x^2 + x - 9}{2x - 8}$

degree on top: _____

degree on bottom: _____

horizontal asymptote:

_____

b. $h(x) = \dfrac{x^2 + 5x + 4}{3x^2 - 12}$

degree on top: _____

degree on bottom: _____

horizontal asymptote:

_____

c. $k(x) = \dfrac{x}{(2x - 1)(x + 6)}$

degree on top: _____

degree on bottom: _____

horizontal asymptote:

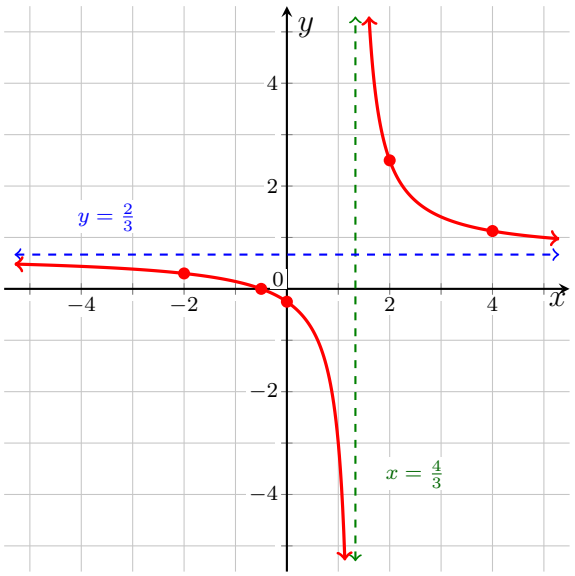
_____

Say it: In part b the asymptote is not $y = 0$ because the degrees are _____, so I use the ratio of the _____.

Example 4: Graph a Function of the Form $\frac{ax+b}{cx+d}$

What is the graph of the function $f(x) = \dfrac{2x + 1}{3x - 4}$?

Step 1 Vertical asymptote: where the **denominator** is 0.
$3x - 4 = 0$, so $3x = 4$, so $x = \frac{4}{3}$.
The numerator is not 0 there ($2 \cdot \frac{4}{3} + 1 = \frac{11}{3}$), so there is a vertical asymptote at $\boldsymbol{x = \frac{4}{3}}$.
Rule (Concept Summary): $x = a$ is a vertical asymptote of $\frac{P(x)}{Q(x)}$ if $Q(a) = 0$ and $P(a) \neq 0$.
Step 2 Horizontal asymptote: the degrees are equal (1 and 1), so use the ratio of the **leading coefficients**: $y = \frac{2}{3}$. As $x \to \pm\infty$, $y \to \frac{2}{3}$.
Horizontal asymptote $\boldsymbol{y = \frac{2}{3}}$.
Step 3 Graph. Draw both asymptotes dashed.
Pick x-values on *both sides* of $x = \frac{4}{3}$ and evaluate:



x	-2	-0.5	0	2	4
$f(x)$	0.3	0	-0.25	2.5	1.125

For example $f(2) = \frac{2(2)+1}{3(2)-4} = \frac{5}{2} = 2.5$. Plot the points and sketch each branch so it bends toward the asymptotes.

Say it: The vertical asymptote is $x =$ _____ because the denominator _____. The horizontal asymptote is $y =$ _____ because the degrees are _____ and the ratio of the leading coefficients is _____.

Try It 4: Graph each function.

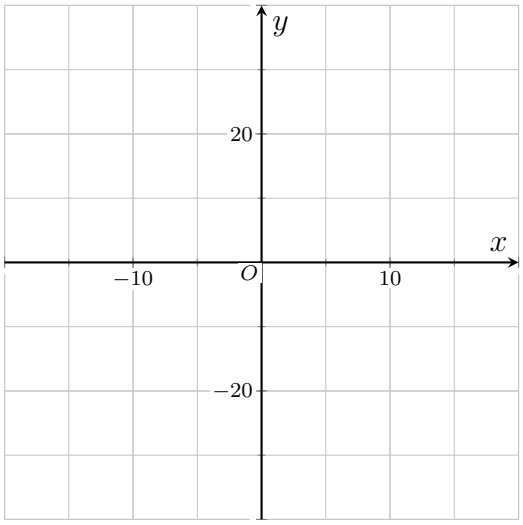
Hint: find both asymptotes, draw them dashed, make a table with x-values on both sides of the vertical asymptote, then sketch.

a. $f(x) = \dfrac{4x-3}{x+8}$

vertical asymptote: $x =$ _____

horizontal asymptote: $y =$ _____

x				
$f(x)$				

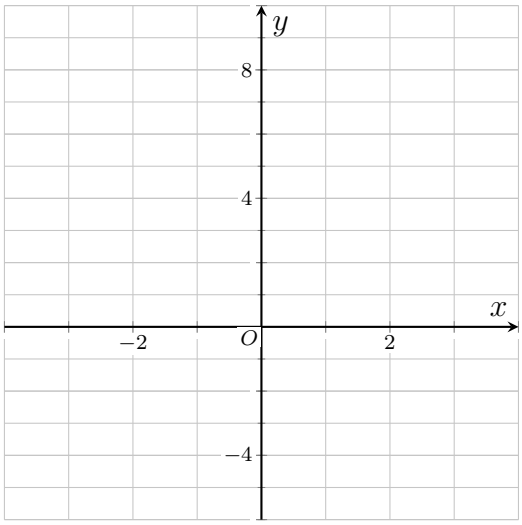


b. $g(x) = \dfrac{3x+2}{x-1}$

vertical asymptote: $x =$ _____

horizontal asymptote: $y =$ _____

x				
$g(x)$				



Practice: Adding water to a mixture

A juice maker adds x liters of water to a tank that holds 5 liters of a drink that is 20% sugar. The fraction of the new mixture that is sugar is $f(x) = \dfrac{1}{x+5}$.

a. Why is the numerator 1?

b. How many liters of water must be added to make a 4% sugar mixture? ($4\% = 0.04$. Solve $\frac{1}{x+5} = 0.04$, or graph $y = f(x)$ and $y = 0.04$ and find where they meet.)

c. What is the horizontal asymptote of f? What does it mean for the drink when a lot of water is added?

d. The vertical asymptote is $x = -5$. Why does it not matter here?

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
7JJ ★	Rational functions: asymptotes and excluded values	Quiz 3 and 5: both asymptotes from the equation (Examples 3 and 4)
79Q ★	Domain of rational functions	Quiz 1 and 3: the denominator is 0 at the vertical asymptote (Example 4, Step 1)
ULD	Domain and range of reciprocal functions	Warm-up for Quiz 1 (from Lesson 4-1)
UB5	Factor quadratics	Warm-up: factor a denominator to find where it is 0 (Exit question; Quiz 2, full Savvas quiz only)

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Describe and correct the error

Juanita is trying to determine the vertical and horizontal asymptotes for the graph of the function $f(x) = \dfrac{x^2 + 3x - 4}{x^2 - x - 12}$. Here is her work:

$$f(x) = \frac{x^2 + 3x - 4}{x^2 - x - 12} = \frac{(x+4)(x-1)}{(x+3)(x-4)}$$

vertical asymptote: $x = -3$, $x = 4$

horizontal asymptote: $y = -4$, $y = 1$ ×

Describe and correct the error Juanita made in determining the vertical and horizontal asymptotes.

Juanita's vertical asymptotes are _____ because the _____ is 0 at $x = -3$ and $x = 4$ and the numerator is not.

Her horizontal asymptotes are wrong because she used _____.

The degree of the numerator is _____ the degree of the denominator, so the horizontal asymptote is $y =$ _____, the ratio of the _____.